Stellar Interferometry at the CHARA Array

Measuring the Sizes of Stars Using Multiple Telescopes:

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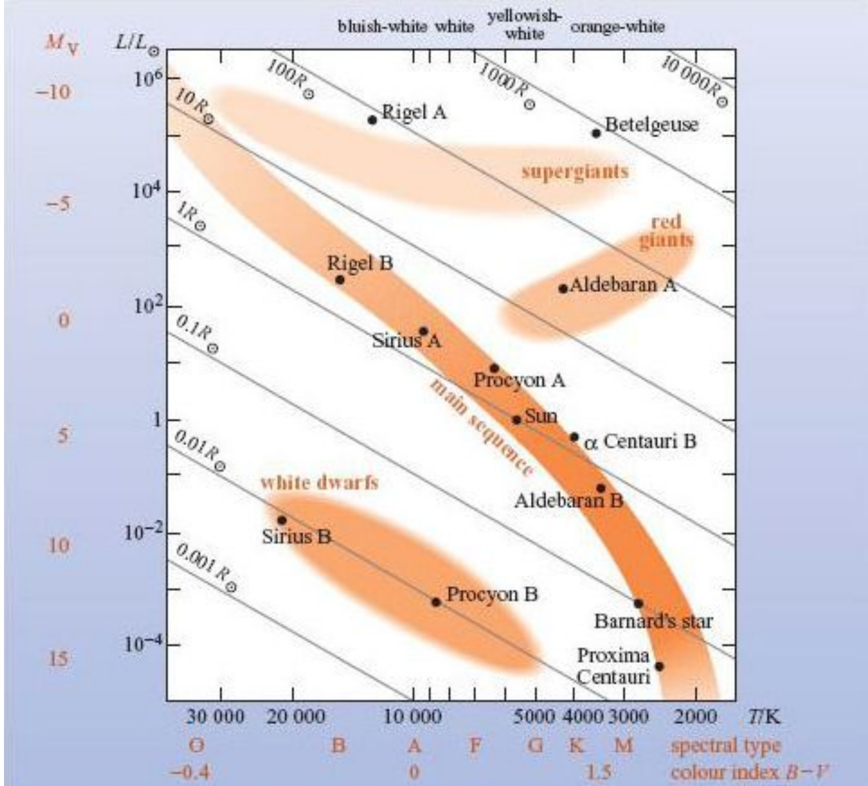
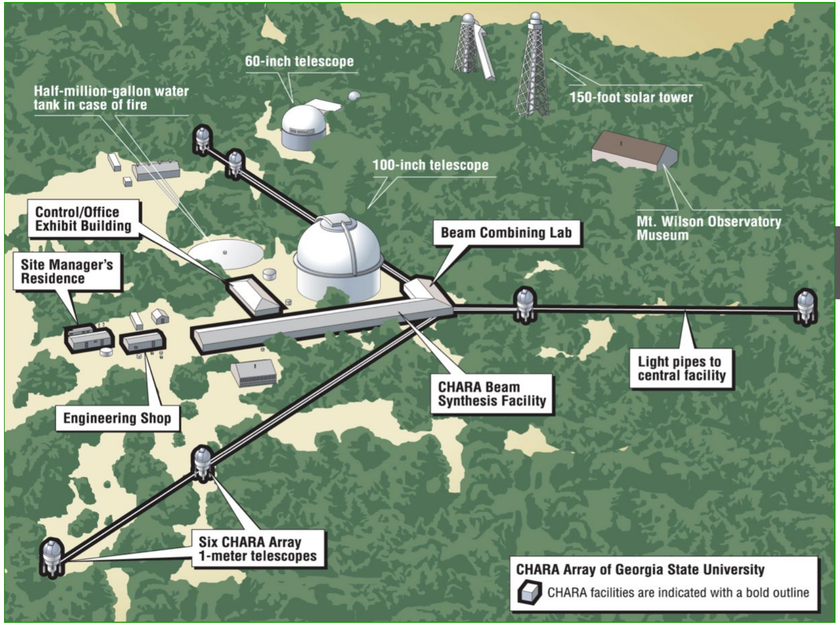
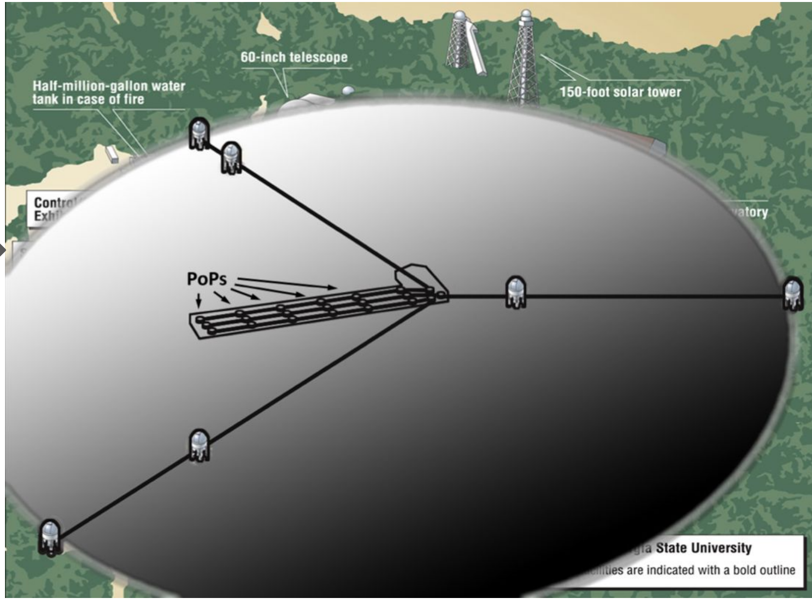
On top of Mount Wilson, where Edwin Hubble made his observations that the universe is expanding..

The Center for High Angular Resolution Astronomy (CHARA) array is a 6 telescope utilizing interferometric infrared observations in order to measure diameters of stars.

- Knowing a star's size helps us understand how much fuel it has and how it will change with time.







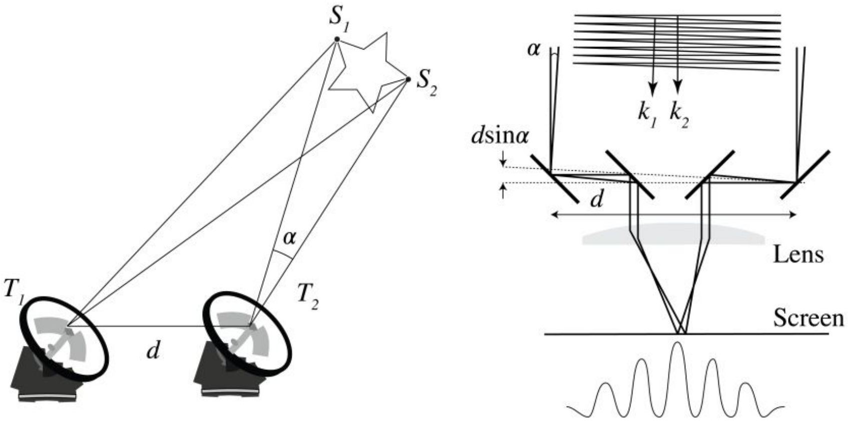
With six telescopes arranged in different pairs and triangles, CHARA acts like one giant "virtual telescope." The sharpness comes from the baseline, the distance between telescopes, which reaches ~330 meters.

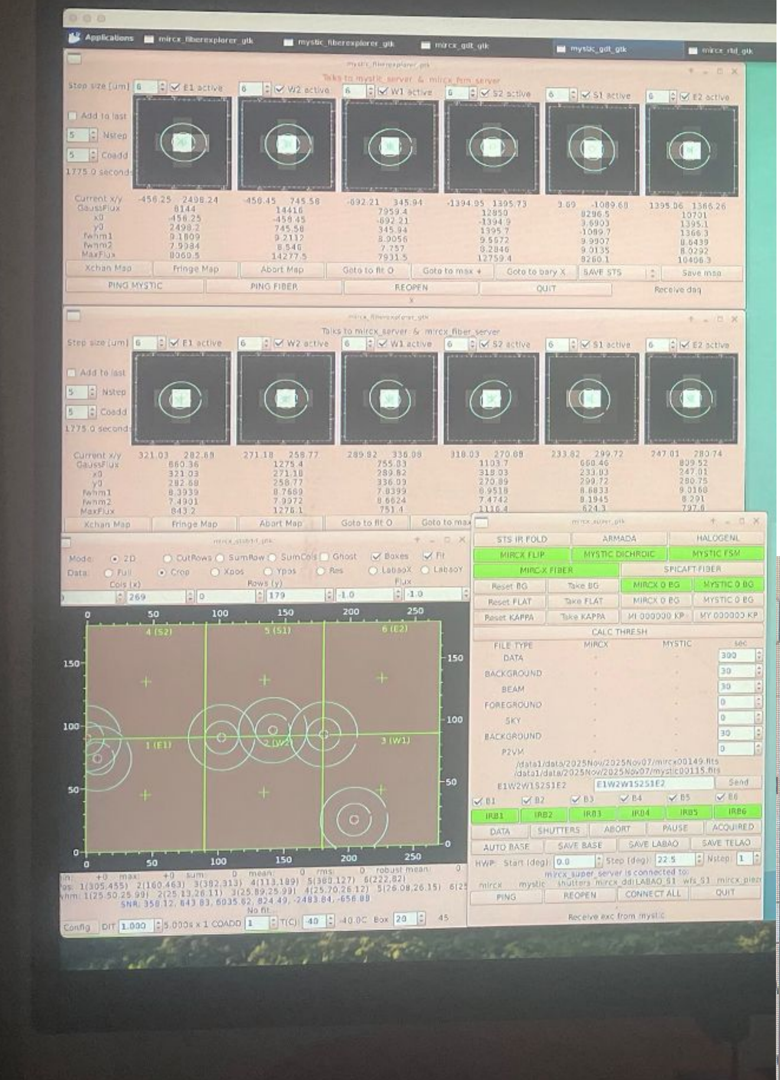
Using infrared light and this huge baseline, CHARA achieves extremely fine angular resolution (like seeing a nickel from ~100,000 miles away), making it ideal for measuring stellar diameters and, in some cases, imaging stellar surfaces.

Think of the light beams like runners from different starting lines who all need to cross the finish line at the same millisecond.

Because the telescopes are at different distances from the star, the light reaches them at different times. To fix this, these motorized delay carts act like flexible "extra track" that can be instantly lengthened or shortened for each runner. By moving continuously along precision rails, they adjust the distance the light travels to ensure every beam arrives at the lab in a perfect "tie," allowing them to interfere and create a high-resolution image.

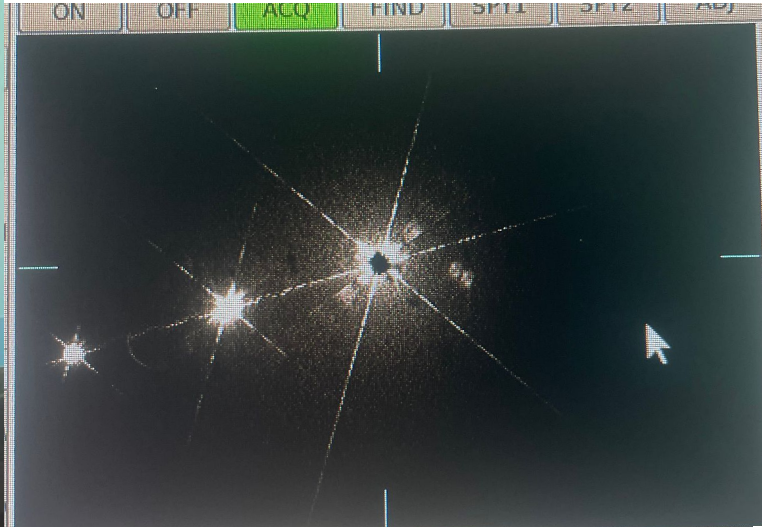
This timing correction lets the light combine properly to form interference patterns (fringes). Once the timing is matched, the observer can ‘lock’ the system and record data.

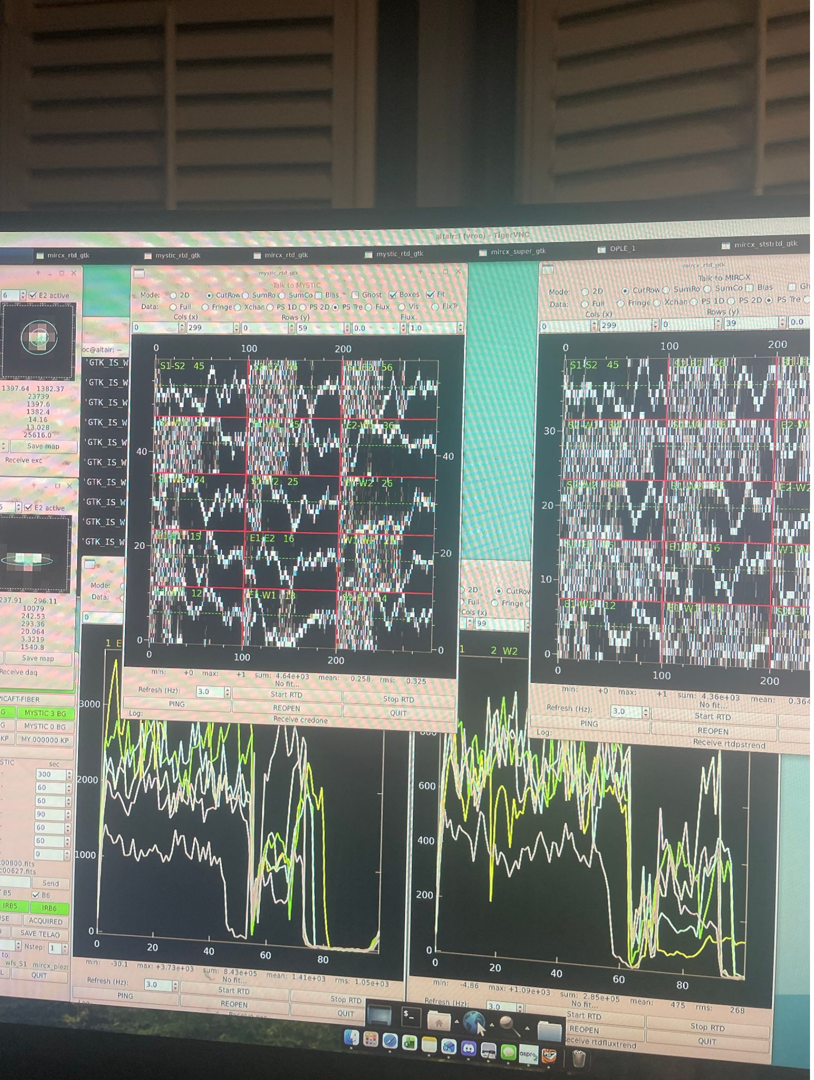




Remotely Observing - 11/6-8 2025 Locking on with each of the 6 telescopes

All telescopes must stay precisely aligned on the same star for interferometry to work.
Wind can knock out one or two telescopes temporarily.
Facilities are closed during rain and forest fires, otherwise running every night of the year.





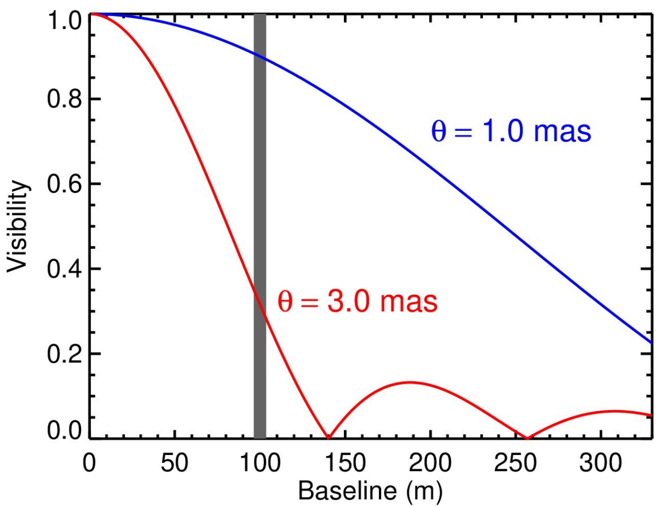
Remotely Observing- Fringe locking and collecting data!

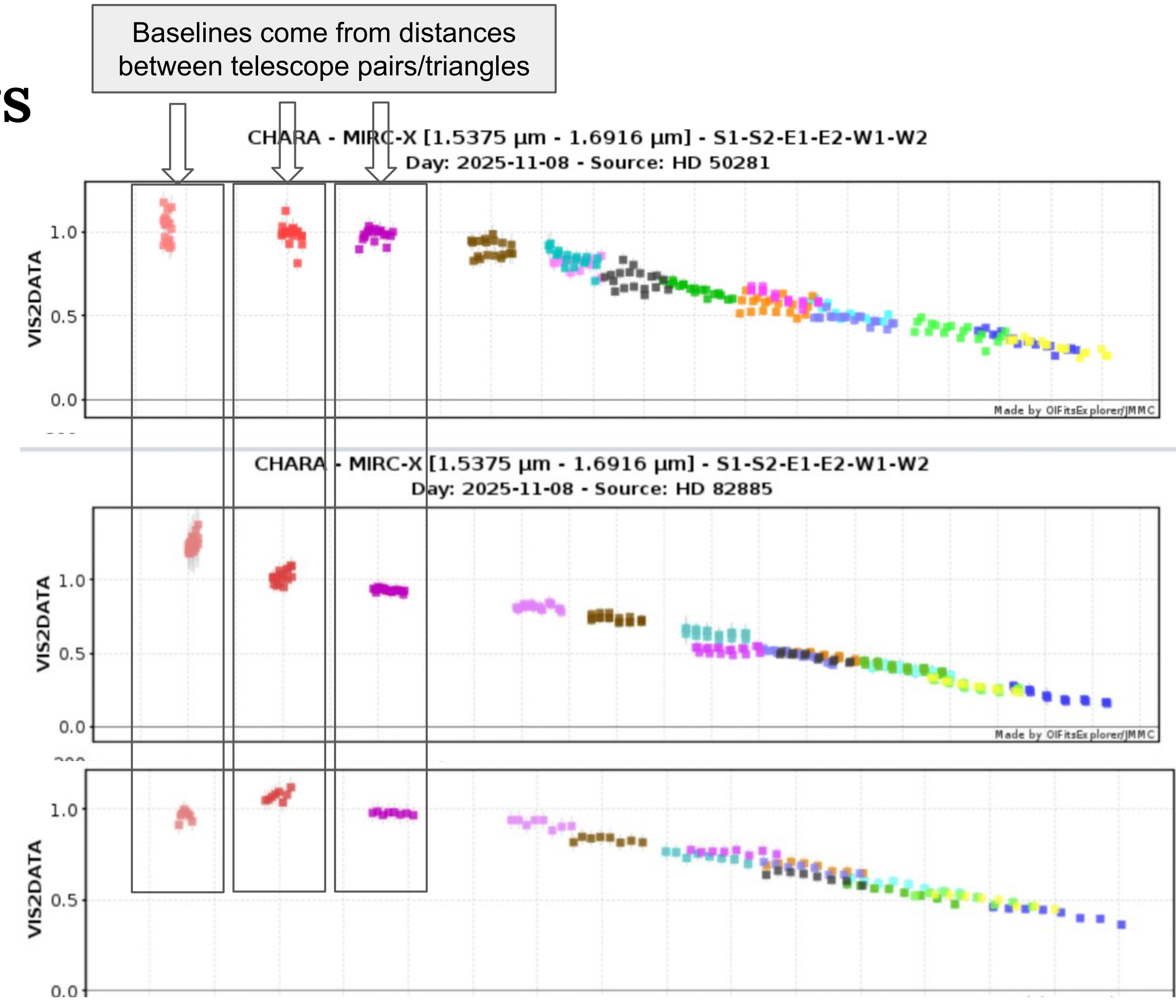
In the top two panels, we lock on the fringes and take measurements with the MIRC-X and MYSTIC instruments at 5 minute cadences. A ‘good’ fringe appears as a centered, smooth line and indicates stable, constructive interference (light arriving together in phase). Noisy or unstable signals indicate poor interference and unusable data.

Observing flux (quantifies light received) in each of the six telescopes for both instruments in the bottom two panels. A loss of flux requires re-aligning the telescope to recover the light.

Measuring diameters

After reducing a full night of data, we calibrate the target star's visibility using reference stars observed before and after it. The visibility decreases as the interferometer resolves the star's edges. Fitting a model (Bessel) curve to this falloff gives the angular diameter.





Measuring inclination and imaging

Using infrared at each point of the star we can deduce temperature and reconstruct images from visibility curves, and measure the incline and geometric warping due to fast rotation.